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Ritvik Kalsotra

Summary

I am an M.Sc. Biotechnology student at Lovely Professional University, Punjab, India, with a strong foundation in structural bioinformatics, molecular modelling, and computational drug discovery.

I recently completed my dissertation at CSIR–Central Drug Research Institute (CDRI), Lucknow, entitled “Integrated machine learning & structure-based approach for identification of selective and potent SGLT2 inhibitors”, and am currently undertaking a skill development program in transcriptomics and bioinformatics tool development at CSIR–Institute of Himalayan Bioresource Technology (IHBT), Palampur. With hands-on experience in structure-based drug discovery, molecular dynamics simulations, and machine learning applied to biological systems, I aim to pursue a PhD in Structural Bioinformatics to contribute to the development of computational approaches for rational drug design.

Areas of Proficiency

Programming & Scripting:

Python, R

Tools & Software:

Linux, Cytoscape, AutoDock Vina, PyMOL, Chimera, AutoDock 4, Modeller, SwissDock, GROMACS, Discovery Studio, Avogadro, RDKit

Methods & Concepts:

Machine learning, Deep learning, Network pharmacogenomics, NGS and Transcriptomics (RNA-seq analysis), Molecular docking and dynamics simulations (MD), Structure-based drug discovery, Homology modelling, Virtual screening, ADMET analysis, Bioinformatics tool and pipeline development

Wet Lab & Molecular Biology Techniques:

DNA and RNA isolation, Agarose gel electrophoresis, PCR and primer handling, Microbial culture techniques, Media preparation and sterilization, Pipetting and solution preparation, Spectrophotometric analysis, Protein estimation techniques, Basic chromatography techniques, Cell culture handling, Biochemical assays, Laboratory safety and GLP practices

Websites And Profiles

LinkedIn: www.linkedin.com/in/ritvik-kalsotra-227959308

Education

M.Sc. Biotechnology, 2024 - 2026
Lovely Professional University, Phagwara, Punjab, India
GPA: SGPA – Sem 1: 8.44, Sem 2: 8.56, Sem 3: 8.79

B.Sc., Biotechnology, 2021 - 2024
University of Himachal Pradesh, Shimla, Himachal Pradesh
Graduated with 1st class honors (CGPA: 6.9)

Experience

Skill Development Program Trainee, CSIR–Institute of Himalayan Bioresource Technology (IHBT), Palampur, India

Ongoing

- Learning transcriptomics data analysis workflows (RNA-seq).
- Developing bioinformatics tools and pipelines under the guidance of Dr. Vishal Acharya, Scientist E.

Dissertation Trainee (6 months), CSIR–Central Drug Research Institute (CDRI), Lucknow, India

January - June 2026

Dissertation Project

1. **AI/ML and Structural Bioinformatics approach to identify a potential SGLT2 inhibitor**
   - Performed structure-based virtual screening of compound libraries against the SGLT2 receptor using AutoDock Vina and AutoDock 4.
   - Conducted molecular dynamics simulations (GROMACS) to assess binding stability and conformational dynamics of lead compounds.
   - Developed ML models integrating physics-based descriptors (docking scores, MM-GBSA energies) for binding affinity prediction.

Conferences

1. **Participant:** "Bioinnovate 2025: Bridging Science and Society for Sustainable Development Goals"
2. **Poster Presentation:** "Exploring the Therapeutic Potentials of Phytochemicals against Type 2 Diabetes" at Bioinnovate 2025: Bridging Science and Society for Sustainable Development Goals

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Workshops & Certifications

- Python for Genomic Data Science – Johns Hopkins University (Coursera)
- Biology Meets Programming: Bioinformatics for Beginners – University of California San Diego (Coursera)
- Bacterial Bioinformatics – University of Virginia (Coursera)
- Workshop on Network Biology and Visualization through Cytoscape and GSEA – BRIC–National Centre for Cell Science (BRIC-NCCS), Pune | May 2026

References

Dr. Mohammad Imran Siddiqi
Scientist G
CSIR–Central Drug Research Institute (CDRI), Lucknow
Relation: Mentor

Dr. Vishal Acharya
Scientist E
CSIR–Institute of Himalayan Bioresource Technology (IHBT), Palampur
Relation: Mentor